

DISENTANGLING MORPHOMIC SPLITS IN LIMBU

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ABSTRACT: We shall examine patterns of syncretism with participant marking in Limbu (van Driem 1987), where verbs are inflected for person/number of core A, P, and S roles. While on the one hand, participant marking in Limbu appears highly transparent, witnessing discrete person and number markers for each role, we find a number of different syncretism patterns that obscure the general transparency. Such syncretisms range from almost full neutralisation of number for both participants ($2 > 1$), via neutralisation of P number ($1_{\text{NON-SG}} > 2$), to neutralisation of the dual/plural distinction for third person A and P, a case of bidirectional divergent syncretism (Stump 2001). A most peculiar syncretism pattern involves the tense/agreement portmanteau *-mʔna*, giving rise to a morphomic split we call “pseudo-Pāṇinian”. In our formal analysis, we show how pseudo-Pāṇinian splits can be assimilated to standard Pāṇinian splits and apply this approach to the treatment of bidirectional syncretism. Essentially, we shall propose that the “dual” marker *-si* is actually a general non-singular marker. We shall show how these instances of competition and syncretism can be formalised in Information-based Morphology (Crysmann & Bonami 2016).

KEYWORDS: poly-personal agreement, formal morphology, morphome, neutralisation, morphological split

1. INTRODUCTION¹

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Abbreviations: A, P, S: agent, patient, intransitive subject; 1, 2, 3: first, second, third person;

The study of syncretism and the paradigm splits it gives rise to assume a central role in inflectional theory, since they enable us to understand (i) how paradigms are internally structured and (ii) what factors contribute to paradigmatic contrast and competition. Furthermore, syncretism sheds a light on how morphological expression corresponds to the morphosyntactic features being realised, ultimately providing arguments for morphology as an autonomous component of grammar, independent of syntax and phonology. Within the morphological literature, several types of splits have been recognised, ranging from natural splits to (unmotivated) morphomic splits, schematically represented in Figure 1.

A B			A B			A B			A B		
α	x	y	α	x	x	α	x	y	α	x	x
β	x	y	β	x	x	β	x	y	β	x	y
γ	x	y	γ	x	z	γ	z	z	γ	y	y

(a) NATURAL (b) PĀṆINIAN (c) PSEUDO-PĀṆINIAN (d) MORPHOMIC

FIGURE 1: SOME TYPES OF SPLITS

Natural splits, the simplest case, are depicted in Figure 1a. Here, opposition of form (x vs. y) neatly corresponds to a featural distinction (A vs. B). Natural splits are easily captured by underspecification (here: for α, β, γ). On the other end of the scale we find morphomic splits (Figure 1d), where there is no straightforward alignment at all between the distribution of forms (x, y) and features (A, B; α, β, γ), or conjunctive feature combinations. With Pāṇinian splits (Bonami 2015), as in Figure 1b, the distribution of x cannot receive a natural featural description by itself, either. However, almost every morphological theory on the market² caters for this rather common situation, considering z a specific override of default x , i.e. an instance of the Elsewhere Condition (Kiparsky 1985) or Pāṇini's Principle. A central property of Pāṇinian splits is proper nesting: the overriding cells are properly contained within the set of cells denoted by the de-

SG, DU, PL, NSG: singular, dual, plural, non-singular (i.e., dual or plural); DE, PE, DI, PI: (first person) dual exclusive, plural exclusive, dual inclusive, plural inclusive; $\sigma > \tau$: agent σ acting on patient τ (equivalent to $\sigma A. \tau P$); PRET, NPRET: preterite, non-preterite; AFF, NEG: affirmative, negative; $\emptyset$: absence of exponent (contrastive, in the case of zero exponence; or non-contrastive, in the case of neutralisation).

² This includes theories as diverse as Lexical Phonology (Kiparsky 1985), Distributed Morphology (Halle & Marantz 1993; Embick & Noyer 2007), Paradigm Function Morphology (Stump 2001), Optimality Theory (Prince & Smolensky 1993), and, of course, Information-based Morphology (Crysmann & Bonami 2016). A-morphous Morphology (Anderson 1992) equally recognises preemption, but effects this not by appeal to the specificity of descriptions, but rather by extrinsic ordering within rule blocks.

fault. In this paper, we argue for a specific type of morphomic split, nicknamed Pseudo-Pāṇinian (Figure 1c) that shows family resemblance with both natural and Pāṇinian splits.

We shall examine patterns of syncretism in the system of participant marking in Limbu (van Driem 1987), a Kiranti language spoken in eastern Nepal by 180,000 people. Similar to other Kiranti languages, such as Athpare (Ebert 1997b) or Camling (Ebert 1997a), Limbu verbs inflect for their core arguments, corresponding to S(ole), A(gent) and P(atient) roles. Participants are marked for number (singular, dual, plural) and person (1,2,3), including an inclusive/exclusive distinction for first person non-singular. Marking is predominantly suffixal, with only a few prefixal markers for number, person, and negation.

On the one hand, participant marking in Limbu appears relatively transparent: in case of combination (in transitive verbs), participants tend to be marked individually. Adding to the transparency, person and number distinctions for each participant are often marked separately by discrete markers. Moreover, the system of participant marking is largely the same across different tenses (non-preterite vs. preterite) or polarity. In keeping with these observations, George van Driem, the primary source on Limbu, describes verb agreement in the language as “elaborate but efficient” and “succinct” (van Driem 1987: 69).

On the other hand, this transparency contrasts with a number of syncretism patterns that affect different parts of the paradigm in different ways (cf. Table 1 for reference). Besides almost complete neutralisation of second person number contrasts in the $2 > 1$ and $1 > 2$ cells³ (see §2.4), we also find partial neutralisation of the dual/plural distinction for third person A and third person P (see §2.3). This syncretism in the third person differs for A and P roles, providing an instance of divergent bidirectional syncretism in the terminology of Stump (2001). The third type of syncretism that complicates the system can be observed with allomorphic variation of person/number markers in different tenses (or polarities), giving rise to what we shall call “pseudo-Pāṇinian” splits (see §2.2).

In a nutshell, pseudo-Pāṇinian splits are a sub-species of morphomic splits that arise whenever an override generalises across a natural split, as represented schematically in Figure 1c. Because the overriding exponent is literally the same across the natural split, standard conceptions of Pāṇini’s Principle will fail, since there is no proper nesting. The aim of this paper is therefore three-fold: (i) show how the recognition of pseudo-Pāṇini is crucial for a description of Limbu agreement marking, (ii) formally relate pseudo-Pāṇinian splits to their “component” splits (natural, Pāṇinian), in order to (iii) further understand the principles of knowledge use underlying inflectional systems. Finally, we demon-

³ We use the (standard) $\sigma > \tau$ notation to denote any participant with person/number features σ acting on a patient with features τ .

strate how the approach behind our analysis of this phenomenon can be employed to make greater sense of other non-canonical phenomena in morphology, such as L-shaped syncretism or exceptional neutralisation of marking.

The paper is structured as follows: Section 2 will provide the presentation of the data and empirical generalisations, whereas Section 3 will contain the discussion of theoretical and formal challenges, as well as their solution. The two sections are organised in a parallel fashion: the phenomena discussed in Sections 2.2, 2.3 and 2.4 will be analysed in Sections 3.2, 3.3 and 3.4, respectively.

2. SYNCRETISM IN LIMBU PARTICIPANT MARKING

2.1 *Overview of Limbu verbal inflection*

Verbs in Limbu primarily inflect along at least four inflectional dimensions: tense (non-preterite/preterite), polarity (affirmative/negative), as well as agreement with up to two participants. Although additional aspect and mood categories do exist, they are formally orthogonal to the previous dimensions, occurring mostly as final suffixes (van Driem 1987: 69). Periphrastic tenses also involve suffixes to the aforementioned primary forms of verbs, while adhortative and imperative forms inflect similarly to the (zero-marked) indicative mood, with some differences. Therefore, we limit ourselves to these four primary dimensions.

As shown in Table 1, transitive verbs inflect for person and number of both A(gent) and P(atient) participants, differentiating three numbers (singular, dual, plural) and three persons. Marking of first person further distinguishes between exclusive and inclusive in both dual and plural. Scenarios where the two participants necessarily overlap (marked with — in the table) are morphologically impossible and must be expressed with a reflexive.

Expression of tense and polarity is performed by means of independent discrete markers, whose allomorphy is almost always phonologically conditioned. Preterite tense is marked with the suffix *-ε*, while negation employs a circumfixal *mε- -nen*. Hence, there is very little overlap between these two dimensions and person/number marking of participants: most agreement markers are constant across different tenses/polarities, although a few cases of allomorphy/fusion can indeed be observed. In Table 1, cells involving a first singular or first plural participant may involve different forms in the negative, or the preterite, cf. e.g. the 1sg > 3sg cell. We have indicated the alternate forms using a two-by-two mini-paradigm in these cells, with tense being distinguished vertically (top: non-preterite, bottom: preterite) and polarity horizontally (left: affirmative, right: negative).

Exponents for agreement with the S(ole) argument of the intransitive

↓ A \ P →	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU	3PL
1SG	—	—	—	—	—	-nɛ	-nɛ-tchi-ŋ	-n-i-ŋ	$\frac{-ʔɛ}{-paŋ}$	$\frac{-ʔɛ-n-chi-n}{-paŋ-si-ŋ}$	→
1DE	—	—	—	—	—	←	-nɛ-tchi-ge	→	-s-u-ge	-s-u-si-ge	→
1PE	—	—	—	—	—	↙	↓	↘	$\frac{-u-m-be}{-mʔna}$	$\frac{-u-m-si-m-be}{-mʔna-si}$	→
1DI	—	—	—	—	—	—	—	—	a- -s-u	a- -s-u-si	→
1PI	—	—	—	—	—	—	—	—	a- -u-m	a- -u-m-si-m	→
2SG	$\frac{kɛ- -ʔɛ}{kɛ- -aŋ}$	↑	↘	—	—	—	—	—	kɛ- -u	kɛ- -u-si	→
2DU	←	a-ge-	→	—	—	—	—	—	kɛ- -s-u	kɛ- -s-u-si	→
2PL	↙	↓	↘	—	—	—	—	—	kɛ- -u-m	kɛ- -u-m-si-m	→
3SG	$\frac{-ʔɛ}{-aŋ}$	-si-ge	-i-ge	a- -si	a-	kɛ-	kɛ- -si	kɛ- -i	-u	-u-si	→
3DU	↑	↑	↑	↑	↑	↑	↑	↑	-s-u	-s-u-si	→
3PL	$\frac{mɛ- -ʔɛ}{mɛ- -aŋ}$	mɛ- -si-ge	mɛ- -i-ge	a-m- -si	a-m-	kɛ-m-	kɛ-m- -si	kɛ-m- -i	mɛ- -u	mɛ- -u-si	→
S →	$\frac{-ʔɛ}{-aŋ} \mid \frac{-ʔɛ}{-paŋ, -aŋ}$	-si-ge	$\frac{-i-ge}{-mʔna}$	a- -si	a-	kɛ-	kɛ- -si	kɛ- -i	∅	-si	mɛ-

 TABLE 1: LIMBU PERSON MARKING^a (BASED ON CONJUGATION LISTS IN VAN DRIEM 1987: 368-374)

^a Cells with allomorphy conditioned by tense or polarity are divided up into 2 by 2 sub-tables, with non-preterite at the top, preterite at the bottom, affirmative on the left and negative on the right. Regular tense and polarity marking has been omitted from the paradigm in the interest of readability (largely reproducing the non-preterite affirmative paradigm).

We use arrows to represent syncretism between adjacent cells, without necessarily implying any directionality. Scenarios that are morphologically impossible in Limbu are invalidated with an em-dash (—).

paradigm (bottom-most line, marked with $S \rightarrow$) very closely follow those found in the transitive paradigm for cells with a third singular agent (which is itself zero-marked; 3sg line). More precisely, agreement for first and second person S arguments is largely realised in the same way as for the corresponding P arguments with third singular subjects. For third person S meanwhile, we find the same exponents as for third person A.

Zooming in on the expression of person and number, we find that these agreement features are often expressed by discrete markers. The prefix *ke-* (or *ge-* word-internally) expresses second person participants across a considerable number of cells, with different roles (A/P/S) or number. Similarly, the prefix *a-*, which encodes participation of both speaker and hearer in the event (in first person inclusive and $2 > 1$ cells)⁴, has a comparably wide distribution across roles and number, and so does its exclusive counterpart, suffixal *-ge/be*. Another recurrent marker is suffixal *-u*, which expresses third person P across all three numbers. Likewise, non-singular number is equally expressed by discrete markers: common exponents are *-s(i)/-tchi* (dual/plural), *me-* (plural/dual), *-m* (plural) and *-i* (plural).

Despite the fact that the inventory of forms provides the necessary prerequisites for a full differentiation of featural values for all participants, as witnessed by example (1), a cursory glance at the full paradigm reveals that we do encounter areas of substantial syncretism: in Table 1, these are indicated by arrows. For instance, third person dual and plural P are systematically syncretic (as reflected by the horizontal arrows in the last column), and so are third person dual and plural A acting on first and second person ($3 > 1/2$).

- (1) *uks -ε -tch -u -si -ge*
 pick PRET DU A 3P NSGP -1EXCL
 ‘We (DE) picked them.’ (van Driem 1987: p. 103)

Syncretism in Limbu is not restricted to contiguous areas, however: one example is the preterite first person plural exclusive marker *-mʔna*, highlighted in bold in Table 1, which is used with S and A participants. Similarly, the non-preterite (*-ʔε*) and preterite (*-(p)aq*) exponents of first person singular are found not only in the intransitive and the transitive with third person co-arguments, but also in interaction with a second singular A. We shall discuss in detail these somewhat challenging syncretism patterns, and the splits they give rise to, in the next section.

⁴ See Loreau Unger (2023) for a full analysis of this marker *a-*, including both standard inclusive person (with the same role), or across participants, as in the $1 > 2$ subparadigm.

2.2 Pseudo-Pāṇinian splits

The first case of syncretism patterns we shall examine is witnessed in at least two places in the paradigm in Table 1: one involving exponents of first plural exclusive across different tenses and roles (cf. Table 2), most notably the preterite agreement portmanteau **-mʔna**, and the other involving first singular marker **-ʔɛ/-aŋ**.⁵ As we shall see, both these splits instantiate the pseudo-Pāṇinian pattern (cf. Table 1c).

Let us start with first plural exclusive: looking at the 1PE column of Table 1 in both the transitive and intransitive paradigms, it appears that **-mʔna** is a portmanteau, exceptionally used in the preterite intransitive paradigm for the otherwise regular combination *-i-ge*, consisting of the exclusive marker *-ge/-be* and the plural marker *-i*. Both exclusive and plural markers are attested in other areas of the full paradigm as well, such as the first exclusive dual and plural (1DE, 1PE) rows (*-ge/-be*) and the 2PL column (*-i*), witnessing a more general distribution when compared to *-mʔna*.

However, if we look at the entire distribution of *-mʔna*, we find it in the 1PE > 3 cells as well, where it acts as the preterite tense allomorph of the suffix combination *-u-m-be*. Again, each of the three constituent markers is fairly general, marking third person P (*-u*), first/second plural A (*-m*), and first person exclusive (*-ge/-be*). For convenience, Table 2 presents a sub-paradigm with *-mʔna* and its relevant competitors.

Role	A > 3SG		S		3SG > P	
	NPRET	PRET	PRET	NPRET	PRET	NPRET
1PE	-u-m-be	-mʔna	-mʔna	-i-ge	-i-ge	-i-ge
2PL	kɛ- -u-m	kɛ- -u-m	kɛ- -i	kɛ- -i	kɛ- -i	kɛ- -i
1PI	a- -u-m	a- -u-m	a- -ɛ	a-	a- -ɛ	a-

TABLE 2: PSEUDO-PĀṆINIAN SPLIT IN THE PLURAL, INVOLVING *-mʔna*

Since *-mʔna* cumulates number and person marking, whereas its competitors do not, it is indeed more specific than its competitors in almost all respects – combining preterite with first person exclusive plural – except for its role specification. I.e., *-mʔna* is used for A or S (= “nominative”) roles, which is actually not more specific than either *-i* (S or P = “absolutive”) or *-m* (A). Thus, even though *-mʔna* has the narrowest distribution in terms of cell count, and even though, under a naive conception, it cumulates more information than its competitors, it

⁵ In this article, we shall focus on the formal treatment of *-mʔna* and its competitors. The analysis for *-ʔɛ/-aŋ* is largely analogous and can be found in Loreau Unger (2023).

is strictly speaking not more specific; that is, given the standard formal notion of Pāṇinian competition, which relies on proper nesting of cells or on proper subsumption of features, see Section 3.2.

The nature of the split is illustrated in Table 2, showing a non-canonical π shape like that of Table 1c: as one can easily discern, the split is neither fully natural (in other words, balanced), nor fully Pāṇinian. Indeed, to be considered a proper Pāṇinian split, the distribution of *-mṛna* should be properly *embedded* within either that of *-i* (light grey, S or P role) or that of *-u-m* (dark grey, A role).⁶ In the limiting case, this split must therefore be considered morphomic. However, merely considering the split as morphomic misses the motivated nature of the split: apart from *-mṛna*, we find a natural balanced split (cf. Figure 1a), defined by participant role, and within each partition of the split (i.e. leaving aside *-mṛna*'s syncretism), we find what looks like a Pāṇinian pattern: i.e., the portmanteau *-mṛna* taking a “bite” out of the distribution of the more general markers (cf. Figure 1b). The sole problem: the override is syncretic in both cases.

We suggest that the paradoxical situation associated with pseudo-Pāṇinian overrides should and can be handled at the level of formal analysis, provided the formalism offers the necessary analytical devices to capture both the generalisation and its individuation to each leg of the natural split at the same time. As we shall show in Section 3.2, inheritance hierarchies over rules are such a device that makes it possible to represent general properties, shared across the overrides, while at the same time making each instance specific enough to function as a bona-fide Pāṇinian override.

Role	2 > P			3 > P			S
Agent	2PL	2DU	2SG	3PL	3DU	3SG	
1SG	a-gε-	a-gε-	kε- -?ε	mε- -?ε	mε- -?ε	-?ε	-?ε
1DE	a-gε-	a-gε-	a-gε-	mε- -si-ge	mε- -si-ge	-si-ge	-si-ge
1PE	a-gε-	a-gε-	a-gε-	mε- -i-ge	mε- -i-ge	-i-ge	-i-ge

TABLE 3: ANOTHER PSEUDO-PĀṆINIAN SPLIT IN THE FIRST PERSON NON-PRETERITE AFFIRMATIVE, INVOLVING *-?ε*

A slightly similar pattern regarding Pāṇinian competition can be observed for the markers *-?ε/-aṇ*, both exponents of first person singular that show allomorphic alternation between non-preterite (*-?ε*) and preterite (*-aṇ*). For clarity,

⁶ In Section 3.2, we shall provide a formal definition of Pāṇinian competition in terms of proper subsumption between feature structure descriptions and highlight how Pseudo-Pāṇinian splits deviate from this.

we limit the following discussion and Table 3 to the non-preterite affirmative, but the same observations essentially carry over to the preterite. To show the peculiar distribution of the first person singular marker, the first three columns of Table 1 have been isolated and rearranged to form Table 3. First person singular $-ʔɛ$ can be found, without exception, in the intransitive paradigm, as well as the $3 > 1$ transitive sub-paradigm: this corresponds to the four rightmost cells in the first row of Table 3. This same marker also occurs in $2 > 1$, but here, its distribution is limited to the $2SG > 1SG$ cell. It so appears that in the $2 > 1$ sub-paradigm, this exponent, which has otherwise a much wider distribution, serves as an override to the uniform realisation a - “inclusive”, found in the remaining 8 cells of this 3-by-3 sub-paradigm (while the prefixes ke -/ ge - always express second person).⁷

The theoretical issue at hand is similar but not fully identical to the one found with $-mʔna$ above: in order to capture the generality of the markers, one will want to describe $-ʔɛ$ -/ $aŋ$ as first singular markers of “absolutive” (= S/P) roles, a description that is true of all cells where these exponents occur. To capture their exact distribution, however, one will need to establish that this general description stands on two legs: a more general one that holds in the absence of a second person participant ($1SG.S$ or $3 > 1SG$ cells), and a highly specific one for the $2SG > 1SG$ cell. This distinction then accounts for the absence of the markers in $2NSG > 1SG$ cells. So again, we need both a general characterisation of the $-ʔɛ$ -/ $aŋ$ marker, in order to capture the syncretism pattern, as well as two more specific instances, to limit the distribution to the appropriate cells. Incidentally, the $2SG > 1SG$ instance will also be specific enough to preempt default a - in the $2 > 1$ sub-paradigm. We like to reiterate that inheritance-based frameworks lend themselves easily towards capturing both the concrete properties of each instance and the properties they share in a natural way.

2.3 Neutralisation of dual/plural

Although number marking in Limbu generally distinguishes dual and plural, there are regions in the paradigm in Table 1 where this distinction is effectively neutralised: most obviously, for third person P participants, $-si$ functions as a mere non-singular marker. Similarly, in the $3 > 1/2$ cells, the dual/plural distinction is equally neutralised for A participants, now featuring $mɛ$ - as the exponent of non-singular third person A. However, in situations where the contrast between dual and plural is maintained, as e.g. for third person intransitive S, $mɛ$ - serves

⁷ Under a simple “first person singular” interpretation for the suffix, this restriction to singular agents in the $2 > 1$ paradigm is somewhat unexpected, given that the form ke -/ $-ʔɛ$ should in principle be perfectly compatible with a non-singular agent as well. Even more so, since the actual form found in these cells (a - ge -) does not overtly express any first person number.

to mark plural, whereas *-si* expresses dual.

As depicted in the two topmost rows of Table 4, the syncretism of *-si* and *mε-* in third person gives rise to a pattern of two interlocking L shapes. One way to picture this situation is in terms of divergent bidirectional syncretism (Stump 2001) where the dual marker takes on expression of plural in the third person P cells, and the plural marker is extended to expression of dual cells in the third person A cells. Another, possibly equally palatable, way is to regard it as an instance of heteroclis, picturing the $A > 3$ and S cells as a mixed subparadigm.

When taken in isolation, we are confronted with what looks like a “balanced” morphomic split. Indeed, neither of the two markers can receive a straightforward natural characterisation, yet none of the two can be considered a default or an override on the basis of a comparison of third person non-singular cells alone.

Role	$A > 1/2$	$A > 3$	S	$3 > P$	$1/2 > P$
3DU	<i>mε-</i>	<i>-si</i>	<i>-si</i>	<i>-si</i>	<i>-si</i>
3PL	<i>mε-</i>	<i>mε-</i>	<i>mε-</i>	<i>-si</i>	<i>-si</i>
2DU	∅	<i>-si</i>	<i>-si</i>	<i>-si</i>	∅
2PL	∅	<i>-m</i>	<i>-i</i>	<i>-i</i>	∅
1DE	<i>-tchi</i>	<i>-si</i>	<i>-si</i>	<i>-si</i>	∅
1PE	<i>-tchi</i>	<i>-m</i>	<i>-i</i>	<i>-i</i>	∅
1DI		<i>-si</i>	<i>-si</i>	<i>-si</i>	
1PI		<i>-m</i>	∅	∅	

TABLE 4: EXPONENTS OF DUAL AND PLURAL FOR DIFFERENT PERSONS (ROWS) AND IN DIFFERENT TYPES OF SCENARIOS (COLUMNS). BLANK CELLS ARE IMPOSSIBLE COMBINATIONS IN LIMBU.

The picture changes, however, once we include the full range of exponents for non-singular number (in the following rows of Table 4). For now, we leave aside the issue of number neutralisation in $1 > 2$ and $2 > 1$ cells, taking it up in the next section. As it turns out, there is no other nonzero marker that uniquely encodes dual, but there are other markers (dotted *-m*, crosshatched *-i*) that specifically encode plural. The undifferentiated non-singular function of *-si* is also found for the agent in $1NSG > 2$ cells. As a consequence, it is safe to regard *-s(i)/-tchi* as a non-singular marker that only gets restricted to dual by virtue of paradigmatic competition with dedicated plural markers. This is in line with the very wide distribution of *-si*: according to van Driem (1987), the alternation between *-si*, *-s* and *-tchi* is in most cases mere phonologically conditioned allomorphy (cf. van Driem 1987 for details).

Under this perspective, *mε-* can be seen as an A/S 3rd person non-singular

override. However, given the L-shape of this marker's distribution, there is no single, purely conjunctive description that will capture all and only the relevant cells. Rather, the general description will only approximate the region in which the marker appears (= third person nominative non-singular). To narrow down the description to the exact distribution, we need to additionally trace the L-shape by means of a disjunction of its horizontal (3_{PL}) and vertical (3 > 1/2) legs. So, again, we want to refer both to the generalisation shared by all cells featuring *mε-*, and to the more specific instances that capture their exact distribution. This problem is similar to what we found with *-mʔna* and *-ʔε/-aŋ*. As we shall see in section 3.3, the formal solution will be entirely parallel to that invoked for Pseudo-Pāṇinian splits, relying on inheritance hierarchies.

2.4 Neutralisation of number

Some particularly drastic neutralisation patterns affect cells with only speech act participants (2 > 1, 1 > 2), as should be apparent from Table 1.

Person marking for 2 > 1 uses a combination of the role-independent markers for “inclusive” (*a-*) and second person (*kε-*) participants, but number marking is surprisingly absent: essentially, there is no morphotactic reason why the prefixal markers found in this sub-paradigm should not combine with the (suffixal) number markers. There is, of course, exceptional marking of number in the 2_{SG} > 1_{SG} cell, as discussed in Section 2.2 above. Thus, 8 out of 9 cells show no number marking at all for either participant, while the 2_{SG} > 1_{SG} cell has overt marking of first singular P only, with second singular A being inferrable by paradigmatic opposition.

A more abstract rendering of neutralisation of number is given in Table 4: in 2 > 1 cells, second person agents acting on first person are unmarked for number (upper left zeros), and so are first person patients acted on by second person (lower right zeros). This stands in stark opposition to the intransitive, where speech act participants enjoy two nonzero number exponents to render the full three-way number contrast of Limbu. These 2 > 1 cells are not only clearly exceptional in Limbu, but also in other Kiranti languages such as Athpare (Ebert 1997b), giving rise to cross-linguistic variation.

Turning to the 1 > 2 sub-paradigm, we find that person marking there is expressed by the portmanteau marker *-nε*, which encodes a first person A acting on second person (which preempts the role-independent person marker *kε-* by virtue of Pāṇini's principle). Number, however, is less straightforward: essentially, there is a split between singular and non-singular A. With first singular A, we find a neat distinction of P number, differentiating singular (∅) from dual (*-tchi*) and plural (*-i*). With non-singular A, however, only the marker *-tchi* appears

throughout, neutralising the dual/plural distinction for this argument. Moreover, in this area ($1_{NSG} > 2$), no exponent for P number is found at all.

In Section 3.4, we shall propose a formal analysis for number neutralisation that will address the neutralisations discussed above by complementing the inventory of rules of exponence with a set of simultaneous constraints on zones of inflectedness. These constraints will define the three subparadigms where exponents of non-singular are licensed, thus implicitly banning the application of number inflection rules in the zones showing neutralisation.

3. FORMAL ANALYSIS

Previous formal analyses of Limbu participant marking have so far largely focused on the phenomenon of affix copying found with the $-n$ and $-m$ markers (Zimmermann 2012; Stump 2022). Stump does provide a grammar fragment for part of the paradigm, but the intransitive and third person agent sub-paradigms are not covered. Thus, the specific issues of syncretism we are confronted with here have so far not been addressed.

3.1 Information-based Morphology

The analysis we propose is couched in terms of Information-based Morphology (= IbM; Crysmann & Bonami 2016), an inferential-realisational theory of inflection based on a typed feature structure formalism familiar from HPSG (Pollard & Sag 1994). For a concise introduction to the framework, see Crysmann (2021).

(2) Morphological well-formedness:

$$word \rightarrow \left[\begin{array}{l} MPH \quad \boxed{e_1} \circ \dots \circ \boxed{e_n} \\ RR \quad \left\{ \left[\begin{array}{l} MPH \quad \boxed{e_1} \\ MUD \quad \boxed{m_1} \\ MS \quad \boxed{0} \end{array} \right], \dots, \left[\begin{array}{l} MPH \quad \boxed{e_n} \\ MUD \quad \boxed{m_n} \\ MS \quad \boxed{0} \end{array} \right] \right\} \\ MS \quad \boxed{0} (\boxed{m_1} \cup \dots \cup \boxed{m_n}) \end{array} \right]$$

The general architecture of IbM is quite straightforward: a morphosyntactic property set (MS) is expressed by a set of realisation rules. Rules themselves are pairings of a set of morphosyntactic properties “under discussion” (MUD) with a list of morphs (MPH), cf. example (3) below. Since rules can pair single and multiple MUD elements with zero to many morphs, IbM implements the many-to-many nature of inflectional morphology already at the most fundamental level. Beside “morphemic” 1 : 1 correspondences (Figure 3c–d: $-i$, $-ge$), the rules in Figure 3 on page 16 contain examples of cumulation (Figure 3a: $-m?na$), mapping 2–3 functions onto 1 form, and overlapping exponence (Figure 3b: $-u-m$), mapping

2 functions onto 2 forms). As an example, the description for this last rule is shown in (3), in anticipation of Figure 3b.

$$(3) \left[\begin{array}{c} \text{MPH} \left\langle \left[\begin{array}{cc} \text{PH} & \text{u} \\ \text{PC} & 4 \end{array} \right], \left[\begin{array}{cc} \text{PH} & \text{m} \\ \text{PC} & 5 \end{array} \right] \right\rangle \\ \text{MUD} \left\{ \left[\begin{array}{cc} a & \\ \text{NUM} & \text{pl} \\ \text{PER} & \text{l-per} \end{array} \right], \left[\begin{array}{cc} p & \\ \text{PER} & 3 \end{array} \right] \right\} \end{array} \right]$$

Morphs are further structured to be pairs of phonological form (PH) and morphotactic position (PC), which makes it easy to separate constraints on shape from morphotactics (Crysmann & Bonami 2016). In addition to the properties a rule expresses (MUD), it can also be constrained on properties expressed by some other rule (but present in the rule’s MS), which permits to account for allomorphic variation.

In order to fully exploit the underspecification techniques offered by a typed feature structure formalism, rules are organised into monotonic multi-dimensional inheritance hierarchies, as illustrated by Figure 5. Hierarchies permit two kinds of generalisation: vertically, properties asserted of the supertype also hold true of its subtypes, which essentially implements monotonic inheritance. Generalisations about classes of rules can therefore be captured by abstracting out shared properties into a supertype. Rule instances, to be used in the licensing of an inflected word, are the leaf types of the hierarchy.

Horizontally, “boxed” dimensions serve to capture systematic alternation by cross-classification: i.e. fully formed rules are derived by systematically combing exactly one leaf type from one dimension with exactly one leaf type from every other dimension. E.g., in Figure 5, the constraints on zones of inflectedness in the INFL dimension are systematically distributed over the exponence rule types in the EXPO dimension.

Finally, IbM embraces Pāṇinian competition as a general principle of morphological organisation, similar to PFM. Competition between leaf rule types is established on the basis of unilateral subsumption (see the discussion on page 15 in Section 3.2 below). Essentially, the more general description and its more specific competitor are made disjoint:

(4) Pāṇinian Competition (PAN)

- a. For any leaf type $t_1[\text{MUD } \mu_1, \text{MS } \sigma]$, $t_2[\text{MUD } \mu_2, \text{MS } \sigma \wedge \tau]$ is a morphological competitor, iff $\mu_1 \cup \text{set} \sqsubseteq \mu_2 \cup \text{set}$.
- b. For any leaf type t_1 with competitor t_2 , expand t_1 ’s MS σ with the negation of t_2 ’s MS $\sigma \wedge \tau$: i.e. $\sigma \wedge \neg(\sigma \wedge \tau)$ which is equivalent to $\sigma \wedge \neg\tau$.

Having laid out the basics of the formalism, we can now proceed towards the analysis of splits in Limbu, and how they can be reconciled with Pāṇini's Principle, using inheritance hierarchies and cross-classification of rules.

3.2 Pseudo-Pāṇinian split

The first syncretism pattern we shall address in our analysis is an instance of what we call a pseudo-Pāṇinian split. More specifically, we shall base our discussion on the portmanteau *-mʔna* and its competitors, namely *-i(-ge)* and *-u-m(-be)*. We shall start by providing a feature structure representation of the cells that *-mʔna* realises and contrast it with descriptions of each of its main competitors. This will serve to understand more precisely why the preemption observed here is not quite of the Pāṇinian kind, and what it takes exactly to relate the pattern at hand to that more familiar type of competition.

Recall from Section 2.2 that *-mʔna* is characterised by its cumulative nature, expressing preterite tense, first person plural exclusive. Moreover, this exponent is syncretic across S participants and A participants acting on 3rd person patients. A maximally general feature structure description of the cells this marker expresses is given in (5a) below, capturing syncretism by underspecification: as represented by the MUD value, *-mʔna* expresses *pret(erite)* tense and first person exclusive (*Iexcl*) plural (*pl*). The specification for participant role *nom* is the generalisation of *s* and *a* roles: see the different types of participants we postulate in Figure 2a. In order to account for the fact that *-mʔna* is barred in the 1 > 2 sub-paradigm, we constrain the morphosyntactic property set not to include a second person specification.

$$\begin{array}{ll}
 (5) \text{ a. } -mʔna & \text{ b. } -m \\
 \left[\begin{array}{l} \text{MUD } \left\{ \left[\begin{array}{l} \text{nom} \\ \text{PER } Iexcl \\ \text{NUM } pl \end{array} \right], \text{pret} \right\} \\ \text{MS } \neg \{ [\text{PER } 2], \dots \} \end{array} \right] & \left[\begin{array}{l} \text{MUD } \left\{ \left[\begin{array}{l} a \\ \text{PER } l-per \\ \text{NUM } pl \end{array} \right] \right\} \end{array} \right] \\
 & \text{ c. } -i \\
 & \left[\begin{array}{l} \text{MUD } \left\{ \left[\begin{array}{l} abs \\ \text{PER } l-per \\ \text{NUM } pl \end{array} \right] \right\} \end{array} \right]
 \end{array}$$

The sample feature structures in (5) also provide feature representations of (some of) the competitors for *-mʔna*, pictured as defaults: i.e. they describe the cells where *-i* and *-m* will surface as exponents of plural, including also the cells that we expect to be preempted by exceptional *-mʔna*. Note that both exponents are only available for first and second persons, which we assume to fall under a common supertype for speech act participants, called *l(ocal)-per*; the associated ontology of person values we posit is found in Figure 2b. In other words, the

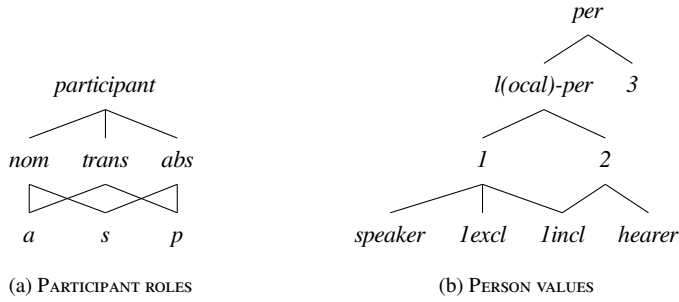


FIGURE 2: TWO TYPE HIERARCHIES: ROLES OF PARTICIPANTS AND PERSON VALUES

feature structures in (5) serve to test how Pāṇinian competition can account for the sub-paradigm in Table 2.

In order to compare the above descriptions regarding the information they contain, i.e. how general or specific they are, we shall apply the standard measure of informativity in feature logic, namely subsumption. Formally, a description d_1 *subsumes* a description d_2 (notated $d_1 \sqsubseteq d_2$) if every feature in d_1 is also a feature in d_2 , and every value in d_1 subsumes every value in d_2 . Since our values are typed, subsumption between values is determined according to the type hierarchy, where a supertype properly subsumes its subtypes (and every type subsumes itself).⁸ Subsumption, as a measure of information in (typed) feature structures, has a systematic relation to the subset relation between denoted objects (here, paradigm cells): $d_1 \sqsubseteq d_2 \leftrightarrow \llbracket d_2 \rrbracket \subseteq \llbracket d_1 \rrbracket$. That is, a morphological feature structure description being more specific, i.e. being properly subsumed ($\sqsubset$), means that the cells it describes are properly embedded ($\subset$) within the cells denoted by the more general, subsuming description.

Under this formal notion of informativity, we can now see explicitly that *-m?na* in (5a) is indeed more specific than *-i* (5c) or *-m* (5b) as regards tense (PRET) and person (1EXCL), and equally specific for number (PL).⁹ However, it falls short when it comes to role: given that it is “nominative” (A or S), the other number markers should be at most as specific as nominative. But *-i* is constrained to an overlapping “absolutive” (S or P) role (cf. Table 2): i.e. in terms of role information, neither subsumes the other. *-m*, by contrast, is in fact strictly more specific in terms of role, restricted to A participants: thus, while *-m?na* is more

⁸ For purposes of exposition, we omit reentrancies, since they do not play a role in the cases we consider here. A complete definition of subsumption will, of course, insist that $d_1 \sqsubseteq d_2$ also requires that every reentrancy in d_1 be a reentrancy in d_2 .

⁹ In order to compare the informativity of sets of MUD values μ_1 and μ_2 , we shall not compare them directly, but rather establish subsumption on the open sets $\mu_1 \cup \text{set}$ vs. $\mu_2 \cup \text{set}$ as in (4). This will permit using subsumption on sets of different cardinality (= size).

specific than *-m* in almost every respect, the inverse holds for role information. This actually reflects the fact that the override (S/A) is not specific to either partition of the natural split (i.e. *-m* (A) vs. *-i* (S/P)), but rather generalises across the split, yielding the π -shaped pattern we regard as characteristic of pseudo-Pāṇinian splits.

Pseudo-Pāṇinian splits do resemble natural and Pāṇinian splits for the very reason that these two well-understood splits are indispensable ingredients. The added ingredient is that the override is identical in form in both parts of the natural splits, a generalisation that needs to be captured. In order to make pseudo-Pāṇinian splits amenable to standard Pāṇinian competition, all it takes is to separate the syncretic aspects of *-mṇa* from its function as an override.

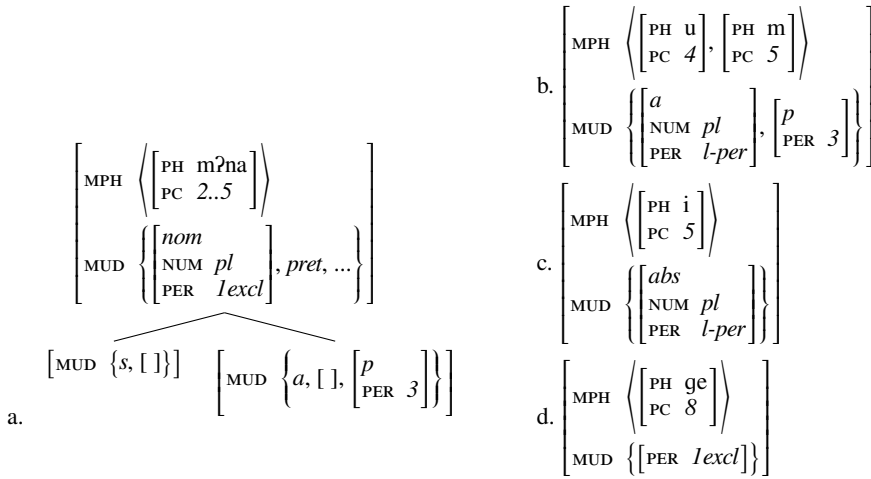


FIGURE 3: *-mṇa* AND ITS COMPETITORS

In IbM, rules are organised in inheritance type hierarchies. As shown in Figure 3, we can capture the division between syncretism of the overrides and their potential in Pāṇinian competition quite straightforwardly, by associating the former with the supertype, yet enumerate as the subtypes the situations in which *-mṇa* functions as an override. Given inheritance, all properties asserted of the supertype are also true of the subtypes. Pāṇinian competition in IbM operates on fully specified leaf types, not on the more general supertype. Therefore, when specialised to the transitive and intransitive uses, it is easy to see that the two instances of generalised *-mṇa* are now properly subsumed by each of its competitors, therefore acting as proper Pāṇinian overrides. Proper subsumption even holds when comparing *-mṇa* to composed competitors such as *-u-m*, which jointly express $1/2\text{PL} > 3$ (top right of Figure 3).¹⁰ As a side effect of enumerat-

¹⁰ As pointed out by Stump (2022), the two exponents *-m* (plural) and *-ṇ* (1sg.A; see the 1sg

ing the use of *-mʔna* in S and A > 3 contexts, we can do away with the negative conditioning constraint on MS in (5a).

Giving the generalised override two more specific legs to stand on enables us to account for all the characteristic features of the split discussed in Section 2.2. First, the naturalness of the split is done justice in that for each of the three “default” markers the descriptions are conjunctive (i.e. not disjunctive), reflecting a contiguous, natural class of cells. Second, the Pāṇinian competition on each side of the split is mirrored with the two specific subtypes of *-mʔna*, carrying just enough information to actually preempt the competitors. Finally, the morphomic-like nature of the split translates to a more complex analysis than the canonical balanced and Pāṇinian splits displayed in Figure 1 of the introduction.

3.3 Neutralisation of dual/plural

The next apparently morphomic split we shall address concerns the realisation of non-singular number that gives rise to a balanced L-shaped split in the sub-paradigms with third person participants.

However, we already observed in §2.3 that the overall distribution of these two markers, *-si* and *mɛ-*, is not balanced at all, with *-si* and its (phonologically conditioned) allomorphs having a much wider share of the full paradigm, cf. Table 4. This is not only true in terms of cell count: indeed some form of *-si* can be found in each of the three persons and each of the three roles (S, A, P). By contrast, *mɛ-* is the only other marker that can have an underspecified non-singular function. Furthermore, we have seen that *-si* does express generic *non-singular* independently of competition with *mɛ-*, e.g. in the 1 > 2 sub-paradigm (see Section 2.3). In fact, whenever *-si* is used with the more specific *dual* function, it always stands in paradigmatic opposition to a specifically marked realisation of plural. This includes overt marking by *-i* (1/2.S/P) and *-m* (1A>3), as well as zero marking for first person inclusive S/P. Thus, we believe it is safe to regard *-s(i)/-tchi* as a default marker for non-singular number.

In order to formally implement this analysis of *-si* as a general non-singular marker, the realisation rules that we propose for this marker, in Figure 4, do not by themselves disambiguate between dual and plural. Rather, disambiguation

row of Table 1) are morphotactically dependent on patient-marking “host” affixes, such as *-u* and *-si*. This dependency is reflected in the dependents’ ability to appear twice (*-u-m-si-m*, in 1/2_{PL} > 3_{NSG} cells), or not at all (*-nɛ* in 1_{SG} > 2_{SG}), according to the number of hosts. In the spirit of Stump’s description, we can capture this phenomenon in IbM by analysing carriers and dependents as orthogonal dimensions of abstraction over rule types contributing two exponents. The *-u-m* rule type in Figure 3 is one example of such a leaf type. See Loreau Unger (2023: §4.4) for details, which involve a minor extension of the formalisation for inflectedness presented in Figure 5.

of number arises by competition and paradigmatic opposition with markers that are either inherently plural (like *-i*, *-m* or plural inclusive *a-* $-\emptyset$), or which are specialised to plural in the relevant cells (*mε-*).

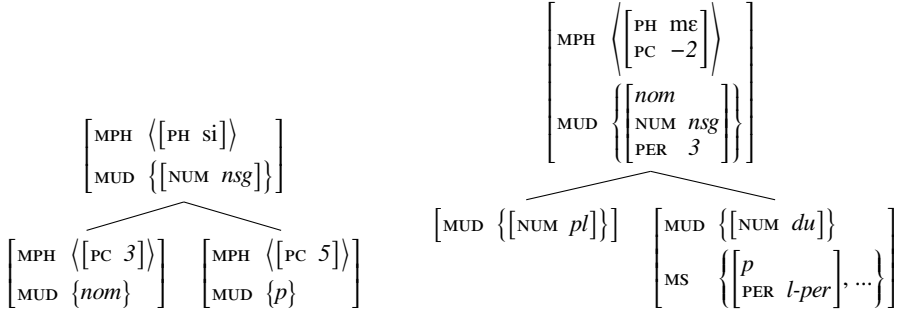


FIGURE 4: *-si* AND *mε-*

Our account of the competition is given in Figure 4: on the left, we see a mini-hierarchy for *-si*. Pretty much all the information we care about in the discussion here is captured by the supertype. The subtypes for *-si* merely constrain the morphotactic position of the marker depending on the participant role it expresses (cf. the simultaneous occurrences of two *-s(i)* markers in *DU > 3NON-SG* cells in Table 1).

Turning now to *mε-*, the supertype gives a minimal description of it as a third person non-singular marker. The two subtypes, however, narrow down the distribution of this marker, to either plural or else dual in $3 > 1/2$ cells. Therefore, even though the exact distribution of *mε-* can only be captured disjunctively, abstracting out a good deal of shared properties will capture syncretism across the disjoint individual rules.

Let us close our discussion of dual/plural neutralisation by inspecting how competition of these two markers will work. The supertype of *mε-* is actually already properly subsumed by the left-hand description for *-si*, which is described as an underspecified non-singular marker for “nominative” S/A roles. Both subtypes for *mε-* add further information with respect to the number value itself (*pl*, *du*), or even a requirement towards the P role to non-third person, changing nothing to their being subsumed by *-si*. Thus, *mε-* will block default non-singular *-si* in these cases, just like the other (equally more specific) exponents of plural, such as *-i* and *-m*.

Taking stock, we have connected the competition between *mε-* and *-si* to the observation that the latter displays characteristic properties of a default non-singular marker, as witnessed by its wide distribution as well as by the presence of other competing number markers that preempt *-si* in a number of plural cells.

As a result, we argue that instead of a morphomic split between two interlocking L shapes (cf. Table 4), what we really find is a Pāṇinian split, where *-si/-tchi* constitutes a default non-singular marker and *mε-* an L-shaped override. Thanks to inheritance, we could capture the syncretism of overriding *mε-* across the disjunctive instances that constitute the L-shape.

3.4 Neutralisation of number: zones of “inflectedness”

Section 2.4 examined the distribution of non-singular markers, specifically their conspicuous absence in “local” configurations, with two speech act participants interacting ($1 > 2$, $2 > 1$). What is crucial about this absence is that it affects all available exponents of number equally: most notably *-si* NSG/DU, *-m* (1/2)PL.A and *-i* (1/2)PL.S/P, including the first person exclusive marker *-ge*.

With the two splits covered in the previous sections, we argued that the reasons behind the morphomic patterns boiled down to competition with other, ultimately more specific markers. In the present case of neutralisation however, the only possible competitor to hypothesise would be a zero override. While this can be a perfectly good solution in certain cases (e.g. for isolated cells like first plural inclusive intransitives), the facts of Limbu suggest otherwise. Indeed, in an area such as $2 > 1$, five different markers could potentially surface: two instances of non-singular *-si* differentiated according to role; plural patient *-i* and plural agent *-m*; and exclusive *-ge*. Therefore, five different override types (one for each combination of number and role, to preempt the corresponding marker, and one for person) would be required to capture the mere absence of any number marking. Because this is far from economical and not very insightful, we suggest instead to address number realisation and neutralisation in terms of zones of “inflectedness”. We reiterate that heavy neutralisation in the $2 > 1$ sub-paradigm can equally be observed in related Athpare (Ebert 1997b).

For the neutralisations at hand, the absence of marking in $2 > 1$ and for patients in $1 > 2$ is the flip side of the *presence* of marking. We identify three distinct, natural zones with overt non-singular marking: (i) the intransitive paradigm; (ii) the “non-local” area of the transitive paradigm, i.e. those cells which involve some third person participant, either A or P; and (iii) the $1 > 2$ subparadigm, i.e. where A is first person and P is second person. These zones are shown abstractly in Table 5. Zone (iii) further subdivides into two sub-areas (not shown): inflection for P number with first singular A (i.e. the neat three-way distinction pointed out at the end of Section 2.3), and neutralisation for P number with first person non-singular A, the latter being marked uniformly by *-tchi*.

Crucially, these three zones license expression of number, independently of the *forms* used for that expression. This is reflected in our formal analysis in

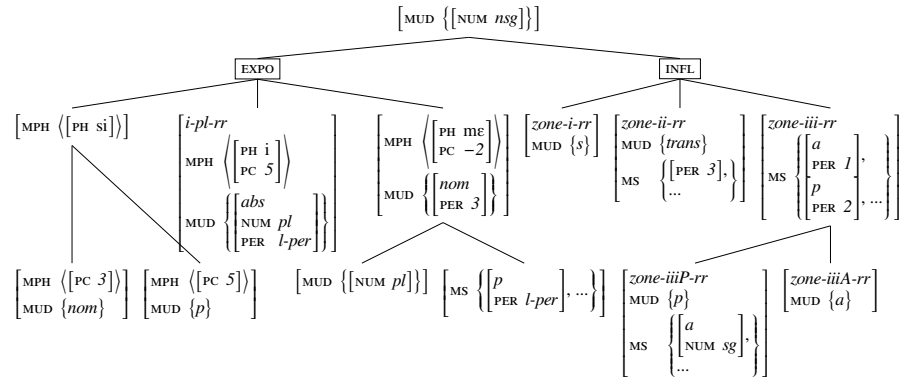
$\downarrow A \setminus P \rightarrow$	1SG	1DE	1PE	1DI	1PI	2SG	2DU	2PL	3SG	3DU	3PL			
1SG	—	—	—	—	—	Zone (iii)								
1DE	—	—	—	—	—									
1PE	—	—	—	—	—									
1DI	—	—	—	—	—	—	—	—						
1PI	—	—	—	—	—	—	—	—						
2SG	(uninflected zone)			—	—	—	—	—						
2DU				—	—	—	—	—						
2PL				—	—	—	—	—						
3SG	Zone (ii)													
3DU														
3PL														
S $\rightarrow$	Zone (i)													

TABLE 5: ZONES OF INFLECTEDNESS AND UNINFLECTEDNESS FOR NUMBER

that we identify areas of inflectedness as a distinction next to the descriptions of exponence, giving two independent, complementary dimensions. In more technical terms, we complement the rules for non-singular number and restrict all these markers' distributions simultaneously by means of a cross-classifying dimension using Online Type Construction (Koenig & Jurafsky 1994; Crysmann & Bonami 2016). This is shown in Figure 5, which recapitulates the relevant number markers from Figures 3 and 4 into a hierarchy under the dimension EXPO, and adds a new dimension INFL. Note that all types inherit from the MUD {[NUM nsg]} specification at the root due to inheritance, thus capturing their nature as exponence of non-singular number. Each of the three zones defined above is described by a top-level type in the INFL dimension (namely *zone-i-rr*, *zone-ii-rr* and *zone-iii-rr*), specifying where the available markers can in fact be used. Once this move is made, no rule type from either dimension can act on its own anymore. Instead, the well-formed leaf rule types (those that actually apply within a word) are now obtained by choosing exactly one leaf type from each of the two dimensions EXPO and INFL and uniting their descriptions.

As can be seen, the EXPO dimension defines the different formal means of marking number, thus providing exponents for the expression of number.¹¹ The INFL dimension, by contrast, makes explicit the three natural zones (i–iii) of non-singular marking. This division of labour between the form-to-function mapping in EXPO and the function-to-form constraints in INFL makes it possible to talk about these two aspects of variation independently. These dimensions are then

¹¹ The person marker *-ge* is left out for reasons of space, but can be integrated straightforwardly; see Loreau Unger (2023: 57).

FIGURE 5: DUAL/PLURAL MARKING (-*si* vs. *mε*- AND -*i*)

combined by cross-classifying each exponence rule with each inflectional zone, giving rise to the final leaf rule types that will be used in any actual word.

This analysis will effectively limit presence of exponents to the subparadigms where number is not neutralised, by way of systematic intersection of leaf types from both dimensions.¹² To see this, consider the rule type *i-pl-rr*: as a member of EXPO, it must combine systematically with the four leaf types in the INFL dimension. In all cases the form information in MPH will be the same, thus capturing the syncretism. But the meaning description will receive specific information, to give several specific instance with different use cases. Three rule types end up being inferred for *-i*, whose meaning descriptions are shown in (6) (form is left out as it is redundant).

- (6) a. *i-pl-rr* & *zone-i-rr*
- $$\left[\text{MUD} \left\{ \begin{bmatrix} s \\ \text{NUM } pl \\ \text{PER } l-per \end{bmatrix} \right\} \right]$$
- b. *i-pl-rr* & *zone-ii-rr*
- $$\left[\text{MUD} \left\{ \begin{bmatrix} abs \\ \text{NUM } pl \\ \text{PER } l-per \end{bmatrix} \right\} \right] \\ \left[\text{MS} \left\{ [\text{PER } 3], \dots \right\} \right]$$
- c. *i-pl-rr* & *zone-iiiP-rr*
- $$\left[\text{MUD} \left\{ \begin{bmatrix} p \\ \text{NUM } pl \\ \text{PER } l-per \end{bmatrix} \right\} \right] \\ \left[\text{MS} \left\{ \begin{bmatrix} a \\ \text{PER } l \\ \text{NUM } sg \end{bmatrix}, \begin{bmatrix} p \\ \text{PER } 2 \end{bmatrix}, \dots \right\} \right]$$

Thus, we license the marker *-i* in all those areas where it is found: e.g., *kε-i* in 2_{PL}.S > (Zone (i)), *-i-qe* in 3_{SG} > 1_{PE} (Zone (ii)), and *-n-i-η* in 1_{SG} > 2_{PL}.

¹² While systematic intersection will of course increase the number of inferrable rule types, the asserted types will remain maximally general.

(Zone (iii)). However, because the type *i-pl-rr* should apply to an absolutive (S or P) participant, it fails to unify with the type *zone-iiiA-rr* which requires an A participant: in case of incompatibility, no rule is inferred. Because of this, we exclude *-i* outside the areas of “inflectedness”: none of the rule types in (6) are available in the uninflected corner $2 > 1$, nor in $1_{\text{NSG}} > 2$, a zone where only number markers for agent (*-si/-tchi* and *-ge*) may appear. This shows how zones of inflectedness can be put into a dimension orthogonal to that used for exponents, yielding systematic uninflectedness of number in the crucial areas outlined in Section 2.4.

4. CONCLUSION

In this paper we have investigated syncretism patterns affecting number marking in Limbu, and we have shown that some of these patterns give rise to morphomic splits that bear too much of a resemblance with both natural and Pāṇinian splits to be treated as ultimately unmotivated. Looking precisely into the behaviour of the tense/agreement portmanteau *-mʔna*, we observed that this marker indeed behaves like a Pāṇinian override for each of its competitors, but the very fact that its form generalises across an otherwise natural split makes application of Pāṇini’s principle fail, since the generalised description of the override is not strictly more specific than those of the forms it preempts. The second morphomic split we investigated concerned neutralisation of the dual/plural contrast with 3rd person participants, giving rise to a double L-shaped split. Showing that the marker *-si* is actually a default exponent of non-singular, we could reduce the split to an L shaped override (*mε-*).

The formal analysis we have proposed, within Information-based Morphology (= IbM; Crysmann & Bonami 2016), crucially exploits inheritance hierarchies of realisation rule types. We have shown that such hierarchies enable us to complement a generalised description of the override, which captures its own syncretism, and provide it with instances that make it just specific enough to work with standard notions of Pāṇinian competition. We successfully applied this technique to both pseudo-Pāṇinian (*-mʔna*) and L-shaped overrides (*mε-*).

The third case of syncretism we studied concerns unexpected full neutralisation of number marking in two sub-paradigms, involving both participants ($2 > 1$) or number marking for the P argument ($1_{\text{NON-SG}} > 2$). We suggested to capture the macroscopic distribution of number marking in terms of zones of “inflectedness” and showed how cross-classifying rule hierarchies in IbM permit to describe these zones in a general fashion, independent of form.

The situation exemplified by *-mʔna*, where an override is itself syncretic across a split, is not an isolated phenomenon of Limbu: we have so far iden-

tified pseudo-Pāṇinian splits in other languages both within Kiranti (Camling, Ebert 1997a; Athpare, Ebert 1997b), as well as in unrelated languages like Koryak (Žukova 1972), Turkana (Dimmendaal 1983) and Pashto (Tegey & Robson 1996).

To conclude, pseudo-Pāṇinian splits and exceptional neutralisation of number marking in Limbu highlight the usefulness of underspecification and cross-classification in hierarchies of inflectional rules, to better encode and reconcile conflicting generalisations in complex morphological systems.

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